

Program: Diploma in Fire Technology and Safety	
Course Code: 5463B	Course Title: Safety in Material Handling
Semester: 5	Credits: 4
Course Category: Program Elective	
Periods per week: 4 (L:4, T:0, P:0)	Periods per semester: 60

Course Objectives:

- To understand the concepts, principles, and significance of material handling in ensuring workplace safety.
- To explore various types of material handling equipment and their correct usage.
- To examine the risks associated with manual material handling (MMH) and mechanical material handling (MMHE).
- To learn safety protocols and techniques to mitigate hazards associated with manual and mechanical material handling operations.
- To incorporate the latest safe methods in material handling operations and understand their practical applications.

Course Pre-requisites:

Topic	Course Code	Course Name	Semester
Principles of Safety Management		Principles of Safety Management	2

Course Outcomes

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Understand the principles and significance of material handling in safety management.	15	Understanding
CO2	Identify and classify various types of material handling equipment and their uses.	14	Applying

CO3	Analyze the risks associated with manual handling and suggest safety measures.	14	Analyzing
CO4	Understand safety considerations and control measures for mechanical material handling.	15	Applying
	Series Test	2	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	2	2				
CO2	3	3	3				
CO3	3	3	2	3			
CO4	3	3	3	3			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive level
CO1	Understand the principles and significance of material handling in safety management.		
M1.01	Material Handling: Concepts and significance	3	Understanding
M1.02	Principles of Material Handling: Objectives and efficiency.	3	Understanding
M1.03	Material Handling Equipments: Types and selection criteria..	4	Applying
M1.04	Systems Concept of Material Handling: Elements and advantages of correct handling.	2	Understanding
M1.05	Limitations and drawbacks in material handling and how to overcome them.	3	Analyzing

Contents: Introduction to Material Handling

Principles and Concepts: Overview of material handling and its importance in ensuring operational efficiency and safety.

Material Handling Equipment: Types and selection of the right equipment for specific handling tasks.

Systems Approach: The systematic integration of equipment, materials, and operators to improve safety and efficiency.

Advancements in Handling Methods: Introduction to new methods such as automation, robotics, and AI integration in material handling systems.

CO2	Identify and classify various types of material handling equipment and their uses.		
M2.01	Classification of Material Handling: Manual, semi-automatic, and automatic systems.	2	Understanding
M2.02	Risk Factors Associated with Manual Handling: Causes and prevention strategies.	3	Analyzing
M2.03	Methods to prevent back injury in manual material handling (MMH).	3	Applying
M2.04	Ways to reduce the risk associated with manual material handling.	3	Applying
M2.05	Emerging Safe Methods in Manual Handling: Use of exoskeletons, powered aids, and ergonomic equipment.	3	Understanding
	Series Test I	1	

Contents: Classification and Risk Factors in Material Handling

Material Handling Systems: Classifying handling systems based on their operation type.

Risk Factors: Examining the common risks associated with manual handling, including back injuries and musculoskeletal disorders.

Preventive Measures: Effective methods to minimize risks during manual handling tasks.

Modern Safety Technologies: Introduction of exoskeletons, powered lifting aids, and ergonomic assistive devices to enhance worker safety.

CO3	Analyze the risks associated with manual handling and suggest safety measures.		
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M3.01	Safety Tips for Manual Material Handling: Proper lifting techniques, posture.	3	Applying
M3.02	Common Safety Considerations: Preventing slips, falls, and strains during manual handling.	4	Applying
M3.03	Ergonomics in Manual Material Handling: Designing workstations to reduce strain.	4	Analyzing
M3.04	Latest Trends in Safety in MMH: Wearable technology and IoT-enabled monitoring systems for risk mitigation.	3	Applying

Contents: Safety Considerations in Manual Material Handling

Safety Techniques: Proper lifting, carrying, and positioning methods to reduce the risk of injury.

Ergonomics: How to design and modify workstations for better safety and comfort in manual handling.

Preventive Practices: Tips for preventing common injuries, such as strains and back injuries.

IoT-enabled Monitoring Systems: The use of smart sensors, wearables, and tracking devices to monitor workers' movements and prevent overexertion.

CO4	Understand safety considerations and control measures for mechanical material handling.		
M 4.01	Mechanical Material Handling Equipment (MMHE): Types, uses, and classification.	3	Applying
M 4.02	Safety Risks in Mechanical Material Handling: Identifying hazards and risks.	3	Analyzing
M 4.03	Control Measures for MMHE Hazards: Risk assessment, safety standards.	3	Applying
M4.04	Safety Features in MMHE: Emergency stop systems, overload protection.	3	Understanding
M4.05	Latest Trends in MMHE Safety: Automation, robotic arms, and AI-driven safety protocols.	3	Applying
	Series Test II	1	

Contents: Mechanical Material Handling and Safety Considerations

Mechanical Equipment: Types of mechanical material handling equipment and their applications in various industries.

Safety Hazards: Identifying and assessing risks associated with mechanical equipment, including electrical, mechanical, and ergonomic hazards.

Control Measures: Implementation of safety measures such as proper training, protective devices, and adherence to safety standards.

Innovative Safety Techniques: Exploring the role of automation and robotics in improving safety during material handling processes.

Text /Reference:

T/R	Book Title/Author
T1	Huseyin K. (Editor), Material Handling Handbook, McGraw-Hill, 2003.
T2	R. D. Green, Ergonomics and Safety in Material Handling, CRC Press, 2005.
R1	James E. Tompkins, Facilities Planning, 4th Edition, Wiley, 2010.
R2	Neumann, William P., Manual Material Handling: Risk Factors and Injury Prevention, Wiley, 2001.
R3	National Safety Council, Accident Prevention Manual for Industrial Operations, 13th Edition, 2015.
R4	Kumar, S., & Singh, D., Advanced Material Handling Systems: Applications of Robotics and Automation, Springer, 2021.